1) a)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & a & a+b & b \\ 0 & c & c+d & d \end{pmatrix} = \begin{pmatrix} 0 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \end{pmatrix}$$

$\therefore a = 1, b = 2, c = 2, d = 1$

So $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

1) b)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 & 1 \\ 0 & 3 & 4 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & a & a+b & b \\ 0 & c & c+d & d \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 & 1 \\ 0 & 3 & 4 & 1 \end{pmatrix}$$

$\therefore a = -1, b = 1, c = 3, d = 1$

So $\begin{pmatrix} -1 & 1 \\ 3 & 1 \end{pmatrix}$

1) c)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 & -3 & -2 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & a & a+b & b \\ 0 & c & c+d & d \end{pmatrix} = \begin{pmatrix} 0 & -1 & -3 & -2 \\ 0 & 1 & 0 & -1 \end{pmatrix}$$

$\therefore a = -1, b = -2, c = 1, d = -1$

So $\begin{pmatrix} -1 & -2 \\ 1 & -1 \end{pmatrix}$

2) a)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 \\ 1 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 2 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} b & 2b & a+2b \\ d & 2d & c+2d \end{pmatrix} = \begin{pmatrix} 1 & 2 & 4 \\ 1 & 2 & 1 \end{pmatrix}$$

$b = 1, d = 1$

$a + 2 \times 1 = 4 \Rightarrow a = 2$

$c + 2 \times 1 = 1 \Rightarrow c = -1$

So $\begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix}$

2) b)

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 2 & 4 \\ 2 & 3 & 0 \end{pmatrix} = \begin{pmatrix} -2 & -2 & 2 \\ 0 & -1 & -2 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2b & 2a+3b & 4a \\ 2d & 2c+3d & 4c \end{pmatrix} = \begin{pmatrix} -2 & -2 & 2 \\ 0 & -1 & -2 \end{pmatrix}$

$4a = 2 \Rightarrow a = \frac{1}{2}$

$2b = -2 \Rightarrow b = -1$

$4c = -2 \Rightarrow c = -\frac{1}{2}$

$2d = 0 \Rightarrow d = 0$

So $\begin{pmatrix} \frac{1}{2} & -1 \\ -\frac{1}{2} & 0 \end{pmatrix}$

2) c)

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 2 & -2 & 2 \\ -2 & 4 & 2 \end{pmatrix} = \begin{pmatrix} -3 & 3 & -3 \\ 3 & -6 & -3 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2a-2b & -2a+4b & 2a+2b \\ 2c-2d & -2c+4d & 2c+2d \end{pmatrix} = \begin{pmatrix} -3 & 3 & -3 \\ 3 & -6 & -3 \end{pmatrix}$

$2a - 2b = -3$ (1)

$-2a + 4b = 3$ (2)

$\therefore a = -\frac{3}{2}$ and $b = 0$

$2c - 2d = 3$ (3)

$-2c + 4d = -6$ (4)

$\therefore c = 0$ and $d = -\frac{3}{2}$

So $\begin{pmatrix} -\frac{3}{2} & 0 \\ 0 & -\frac{3}{2} \end{pmatrix}$

3) a)

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

So $(2, 3)$

3) b)

$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$

So $(4, 9)$

3) c)

$\begin{pmatrix}1 & 3\\0 & 1\end{pmatrix}\begin{pmatrix}2\\3\end{pmatrix}=\begin{pmatrix}11\\3\end{pmatrix}$

So $(11,3)$

3) d)

$\begin{pmatrix}1 & 0\\5 & 1\end{pmatrix}\begin{pmatrix}2\\3\end{pmatrix}=\begin{pmatrix}2\\13\end{pmatrix}$

So $(2,13)$

3) e)

$\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}\begin{pmatrix}2\\3\end{pmatrix}=\begin{pmatrix}5\\5\end{pmatrix}$

So $(5,5)$

3) f)

$\begin{pmatrix}-1 & 2\\3 & -4\end{pmatrix}\begin{pmatrix}2\\3\end{pmatrix}=\begin{pmatrix}4\\-6\end{pmatrix}$

So $(4,-6)$

4) a)

This is the identity matrix. All points get mapped to themselves.

4) b)

This is the zero matrix. All points get mapped to the origin, $(0,0)$

5) a)

$\begin{pmatrix}2 & 2\\2 & 2\end{pmatrix}\begin{pmatrix}-3 & 2\\-5 & -1\end{pmatrix}=\begin{pmatrix}-16 & 2\\-16 & 2\end{pmatrix}$

So $A'(-16,-16)$ and $B'(2,2)$

5) b)

$\begin{pmatrix}-3 & 1\\-4 & 2\end{pmatrix}\begin{pmatrix}-3 & 2\\-5 & -1\end{pmatrix}=\begin{pmatrix}4 & -7\\2 & -10\end{pmatrix}$

So $A'(4,2)$ and $B'(-7,-10)$

5) c)

$\begin{pmatrix}1.2 & 3.6\\-2.8 & -1.9\end{pmatrix}\begin{pmatrix}-3 & 2\\-5 & -1\end{pmatrix}=\begin{pmatrix}-21.6 & -1.2\\17.9 & -3.7\end{pmatrix}$

So $A'(-21.6,17.9)$ and $B'(-1.2,-3.7)$

5) d)

$\begin{pmatrix}0 & 0\\ 0 & 7.7\end{pmatrix}\begin{pmatrix}-3 & 2\\ -5 & -1\end{pmatrix} = \begin{pmatrix}0 & 0\\ -38.5 & -7.7\end{pmatrix}$

So $A'(0,-38.5)$ and $B'(0,-7.7)$

6) a)

$PQ = \sqrt{(4-2)^2 + (-3-1)^2} = \sqrt{20} = 2\sqrt{5}$

6) b)

$\begin{pmatrix}5 & -3\\ 4 & -8\end{pmatrix}\begin{pmatrix}1 & -3\\ 2 & 4\end{pmatrix} = \begin{pmatrix}-1 & -27\\ -12 & -44\end{pmatrix}$

So $P'(-1,-12)$ and $Q'(-27,-44)$

$P'Q' = \sqrt{\left(-44-(-12)\right)^2 + \left(-27-(-1)\right)^2} = \sqrt{1700} = 10\sqrt{17}$

7)

$\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}4 & -2\\ -1 & -3\end{pmatrix} = \begin{pmatrix}7 & -7\\ 1 & -11\end{pmatrix} \Rightarrow \begin{pmatrix}4a-b & -2a-3b\\ 4c-d & -2c-3d\end{pmatrix} = \begin{pmatrix}7 & -7\\ 1 & -11\end{pmatrix}$

$4a - b = 7$ (1)

$-2a - 3b = -7$ (2)

$\therefore a = 2$ and $b = 1$

$4c - d = 1$ (3)

$-2c - 3d = -11$ (4)

$\therefore c = 1$ and $d = 3$

So $\begin{pmatrix}2 & 1\\ 1 & 3\end{pmatrix}$

8)

$\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}5 & 2\\ 6 & -1\end{pmatrix} = \begin{pmatrix}-37 & -8\\ 50 & 3\end{pmatrix} \Rightarrow \begin{pmatrix}5a+6b & 2a-b\\ 5c+6d & 2c-d\end{pmatrix} = \begin{pmatrix}-37 & -8\\ 50 & 3\end{pmatrix}$

$5a + 6b = -37$ (1)

$2a - b = -8$ (2)

$\therefore a = -5$ and $b = -2$

$5c + 6d = 50$ (3)

$2c - d = 3$ (4)

$\therefore c = 4$ and $d = 5$

So $\begin{pmatrix}-5 & -2\\ 4 & 5\end{pmatrix}$

9)

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \Rightarrow \begin{pmatrix} a+2b & a+b \\ c+2d & c+d \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$

$a + 2b = 1$ (1)

$a + b = 1$ (2)

$\therefore a = 1$ and $b = 0$

$c + 2d = 1$ (3)

$c + d = 2$ (4)

$\therefore c = 3$ and $d = -1$

So $\begin{pmatrix} 1 & 0 \\ 3 & -1 \end{pmatrix}$

10)

$\begin{pmatrix} 6 & 8 \\ -2 & 9 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 46 \\ 78 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 6a+8b \\ -2a+9b \end{pmatrix} = \begin{pmatrix} 46 \\ 78 \end{pmatrix}$

$6a + 8b = 46$ (1)

$-2a + 9b = 78$ (2)

Solve (1) and (2) to give $a = -3$ and $b = 8$

11)

$\begin{pmatrix} 4 & 10 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} 8 & 76 \\ -5 & 13 \end{pmatrix} \Rightarrow \begin{pmatrix} 4a+10b & 4c+10d \\ 3a+2b & 3c+2d \end{pmatrix} = \begin{pmatrix} 8 & 76 \\ -5 & 13 \end{pmatrix}$

$4a + 10b = 8$ (1)

$3a + 2b = -5$ (2)

$\therefore a = -3$ and $b = 2$

$4c + 10d = 76$ (3)

$3c + 2d = 13$ (4)

$\therefore c = -1$ and $d = 8$

So $\begin{pmatrix} -3 & -1 \\ 2 & 8 \end{pmatrix}$

$a = -3$, $b = 2$, $c = -1$, $d = 8$

12)

$\begin{pmatrix} 3 & k \\ -4 & 2k \end{pmatrix}\begin{pmatrix} a \\ 16 \end{pmatrix} = \begin{pmatrix} -23 \\ -236 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 3a + 16k \\ -4a + 32k \end{pmatrix} = \begin{pmatrix} -23 \\ -236 \end{pmatrix}$

$3a + 16k = -23$ (1)

$-4a + 32k = -236$ (2)

Solve (1) and (2) to give $k = -5$ and $a = 19$

13)

$\begin{pmatrix} 4k & -5k \\ 8k & 3k \end{pmatrix}\begin{pmatrix} k \\ 8 \end{pmatrix} = \begin{pmatrix} -52k \\ 0 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 4k^2 - 40k \\ 8k^2 + 24k \end{pmatrix} = \begin{pmatrix} -52k \\ 0 \end{pmatrix}$

$4k^2 - 40k = -52k \Rightarrow 4k^2 + 12k = 0 \Rightarrow k^2 + 3k = 0$

$\Rightarrow k(k + 3) = 0 \Rightarrow k = 0$ or $k = -3$

$8k^2 + 24k = 0 \Rightarrow k^2 + 3k = 0 \Rightarrow k(k + 3) = 0 \Rightarrow k = 0$ or $k = -3$

As $k \neq 0$, we must have that $k = -3$

14)

$\begin{pmatrix} k + 3 & k - 5 \\ k - 7 & k + 1 \end{pmatrix}\begin{pmatrix} 3k \\ k - 6 \end{pmatrix} = \begin{pmatrix} 44k + 8 \\ 17k + 5 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} (k + 3) \times 3k + (k - 5) \times (k - 6) \\ (k - 7) \times 3k + (k + 1) \times (k - 6) \end{pmatrix} = \begin{pmatrix} 44k + 8 \\ 17k + 5 \end{pmatrix}$

$3k^2 + 9k + k^2 - 11k + 30 = 44k + 8 \Rightarrow 4k^2 - 46k + 22 = 0$

$\Rightarrow 2k^2 - 23k + 11 = 0 \Rightarrow (2k - 1)(k - 11) = 0 \Rightarrow k = \frac{1}{2}$ or $k = 11$

$3k^2 - 21k + k^2 - 5k - 6 = 17k + 5 \Rightarrow 4k^2 - 43k - 11 = 0$

$\Rightarrow (4k + 1)(k - 11) = 0 \Rightarrow k = -\frac{1}{4}$ or $k = 11$

$\therefore k = 11$

15)

$\begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ k \end{pmatrix} = \begin{pmatrix} 2k+3 \\ k-1 \end{pmatrix}$

$\begin{pmatrix} 3 & 2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} k \\ 1 \end{pmatrix} = \begin{pmatrix} 3k+2 \\ 1-k \end{pmatrix}$

So $P'(2k+3, k-1)$ and $Q'(3k+2, 1-k)$

$P'Q' = 2\sqrt{5} \Rightarrow \big((3k+2)-(2k+3)\big)^2 + \big((1-k)-(k-1)\big)^2 = \big(2\sqrt{5}\big)^2$

$\Rightarrow (k-1)^2 + (2-2k)^2 = 20$

$\Rightarrow k^2 - 2k + 1 + 4 - 8k + 4k^2 = 20$

$\Rightarrow 5k^2 - 10k - 15 = 0$

$\Rightarrow k^2 - 2k - 3 = 0$

$(k-3)(k+1) = 0$

$\Rightarrow k = -1$ or $k = 3$

16)

Choose two points, $(1, 0)$ and $(0, 1)$ for example, and find their images:

$\begin{pmatrix} 8 & -6 \\ 4 & -3 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$

$\begin{pmatrix} 8 & -6 \\ 4 & -3 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ -3 \end{pmatrix}$

The equation of the line passing through $(8, 4)$ and $(-6, -3)$ is:

$y - 4 = \frac{-3-4}{-6-8}(x-8) \Rightarrow y - 4 = \frac{1}{2}(x-8) \Rightarrow y = \frac{1}{2}x$

17)

$\begin{pmatrix}1 & -3\\ 2 & 1\end{pmatrix}\begin{pmatrix}x\\ x\end{pmatrix}=\begin{pmatrix}X\\ aX\end{pmatrix}$

$\Rightarrow \begin{pmatrix}x-3x\\ 2x+x\end{pmatrix}=\begin{pmatrix}X\\ aX\end{pmatrix}$

$\Rightarrow \begin{pmatrix}-2x\\ 3x\end{pmatrix}=\begin{pmatrix}X\\ aX\end{pmatrix}$

$X=-2x$

$aX=3x$

$\therefore a(-2x)=3x$

$\Rightarrow a=-\frac{3}{2}$

The image of the line is $y=-\frac{3}{2}x$, so $a=-\frac{3}{2}$

Alternative Method:

Find the image of two points that lie on the line $y=x$:

$\begin{pmatrix}1 & -3\\ 2 & 1\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix}=\begin{pmatrix}-2\\ 3\end{pmatrix}$

$\begin{pmatrix}1 & -3\\ 2 & 1\end{pmatrix}\begin{pmatrix}2\\ 2\end{pmatrix}=\begin{pmatrix}-4\\ 6\end{pmatrix}$

Find the equation of the line that passes through $(-2,3)$ and $(-4,6)$:

$m=\frac{6-3}{-4-(-2)}=\frac{3}{-2}=-\frac{3}{2}$

$\therefore y-3=-\frac{3}{2}(x+2)\Rightarrow y=-\frac{3}{2}x$, so $a=-\frac{3}{2}$

18)

$\begin{pmatrix} -2 & -5 \\ 8 & -3 \end{pmatrix}\begin{pmatrix} x \\ x-2 \end{pmatrix} = \begin{pmatrix} X \\ aX+b \end{pmatrix}$

$\Rightarrow \begin{pmatrix} -2x-5(x-2) \\ 8x-3(x-2) \end{pmatrix} = \begin{pmatrix} X \\ aX+b \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 10-7x \\ 6+5x \end{pmatrix} = \begin{pmatrix} X \\ aX+b \end{pmatrix}$

$X = 10-7x$

$aX+b = 6+5x$

$\therefore a(10-7x)+b = 6+5x$

$\Rightarrow 10a+b-7ax = 6+5x$

$10a+b = 6$ (1)

$-7a = 5$ (2)

Solve (1) and (2) to get $a = -\frac{5}{7}$ and $b = \frac{92}{7}$

The image of the line is $y = -\frac{5}{7}x + \frac{92}{7}$

So $a = -\frac{5}{7}$ and $b = \frac{92}{7}$

Alternative Method:

Find the image of two points that lie on the line $y = x-2$:

$\begin{pmatrix} -2 & -5 \\ 8 & -3 \end{pmatrix}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 16 \end{pmatrix}$

$\begin{pmatrix} -2 & -5 \\ 8 & -3 \end{pmatrix}\begin{pmatrix} 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 10 \\ 6 \end{pmatrix}$

Find the equation of the line that passes through $(-4, 16)$ and $(10, 6)$:

$m = \frac{16-6}{-4-10} = \frac{10}{-14} = -\frac{5}{7}$

$\therefore y-16 = -\frac{5}{7}(x+4) \Rightarrow y = -\frac{5}{7}x + \frac{92}{7}$

So $a = -\frac{5}{7}$ and $b = \frac{92}{7}$

19)

$\begin{pmatrix} a & 3 \\ 2 & b \end{pmatrix}\begin{pmatrix} x \\ 2x+1 \end{pmatrix} = \begin{pmatrix} X \\ 3X+2 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} ax+3(2x+1) \\ 2x+b(2x+1) \end{pmatrix} = \begin{pmatrix} X \\ 3X+2 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} (a+6)x+3 \\ (2+2b)x+b \end{pmatrix} = \begin{pmatrix} X \\ 3X+2 \end{pmatrix}$

$X = (a+6)x+3$

$3X+2 = (2+2b)x+b$

$\therefore 3\big((a+6)x+3\big)+2 = (2+2b)x+b$

$\Rightarrow (3a+18)x+11 = (2+2b)x+b$

$3a+18 = 2+2b$ (1)

$11 = b$ (2)

Substitute (2) into (1):

$3a+18 = 2+2(11) \Rightarrow a = 2$

$\therefore a = 2$ and $b = 11$

Alternative Method:

Find the image of two points that lie on the line $y = 2x+1$:

$\begin{pmatrix} a & 3 \\ 2 & b \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ b \end{pmatrix}$

$\begin{pmatrix} a & 3 \\ 2 & b \end{pmatrix}\begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} a+9 \\ 2+3b \end{pmatrix}$

Substitute $(3, b)$ and $(a+9, 2+3b)$ into $y = 3x+2$:

$(3, b) \Rightarrow b = 3(3)+2 \Rightarrow b = 11$

$(a+9, 2+3b) \Rightarrow 2+3b = 3(a+9)+2$

$\Rightarrow 2+3(11) = 3a+29 \Rightarrow a = 2$

$\therefore a = 2$ and $b = 11$

20)

$(4, 0)$ is the centre of the circle.

$(2, 2)$ lies on the circle.

$\begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix}\begin{pmatrix} 4 & 2 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -6 \\ 12 & 6 \end{pmatrix}$

So C' has centre $(0, 12)$ and passes through $(-6, 6)$

$r' = \sqrt{(-6-0)^2+(6-12)^2} = \sqrt{72}$

$\therefore C'$ has equation $x^2+(y-12)^2 = 72$

21)

$\begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}\begin{pmatrix} x \\ x^2+2x \end{pmatrix} = \begin{pmatrix} X \\ aX^2+bX+c \end{pmatrix}$

$\Rightarrow \begin{pmatrix} -2x \\ -3x^2-6x \end{pmatrix} = \begin{pmatrix} X \\ aX^2+bX+c \end{pmatrix}$

$X = -2x$

$aX^2+bX+c = -3x^2-6x$

$\therefore a(-2x)^2+b(-2x)+c = -3x^2-6x$

$\Rightarrow 4ax^2-2bx+c = -3x^2-6x$

$4a = -3 \Rightarrow a = -\frac{3}{4}$

$-2b = -6 \Rightarrow b = 3$

$c = 0$

The image of the curve is $y = -\frac{3}{4}x^2+3x$

So $a = -\frac{3}{4}$, $b = 3$ and $c = 0$

22)

$\begin{pmatrix} -2 & -2 \\ 6 & -6 \end{pmatrix}\begin{pmatrix} x \\ \frac{1}{x} \end{pmatrix} = \begin{pmatrix} X \\ \sqrt{\frac{a}{b}X^2-\frac{1}{b}} \end{pmatrix}$

$\Rightarrow \begin{pmatrix} -2x-\frac{2}{x} \\ 6x-\frac{6}{x} \end{pmatrix} = \begin{pmatrix} X \\ \sqrt{\frac{a}{b}X^2-\frac{1}{b}} \end{pmatrix}$

$X = -2x-\frac{2}{x}$

$\sqrt{\frac{a}{b}X^2-\frac{1}{b}} = 6x-\frac{6}{x}$

$\therefore \sqrt{\frac{a}{b}\left(-2x-\frac{2}{x}\right)^2-\frac{1}{b}} = 6x-\frac{6}{x}$

$\Rightarrow \frac{a}{b}\left(-2x-\frac{2}{x}\right)^2-\frac{1}{b} = \left(6x-\frac{6}{x}\right)^2$

$\Rightarrow a\left(4x^2+8+\frac{4}{x^2}\right)-1 = b\left(36x^2-72+\frac{36}{x^2}\right)$

$\Rightarrow 4ax^2+\frac{4a}{x^2}+8a-1 = 36bx^2+\frac{36b}{x^2}-72b$

$4a = 36b \Rightarrow a-9b = 0$ (1)

$8a-1 = -72b \Rightarrow 8a+72b = 1$ (2)

Solve (1) and (2) to get $a = \frac{1}{16}$ and $b = \frac{1}{144}$

The image of the curve is $\frac{1}{16}x^2-\frac{1}{144}b^2 = 1$

So $a = \frac{1}{16}$ and $b = \frac{1}{144}$